



TRACKHIRE

Manage Every Step of Your Job Search in One Place

TECH STACK



MongoDB

ex

Express.js



React.js



Node.js



VERSION 1.0 (MVP)



MVP COMPLETED



PROJECT DOCUMENTATION
AUGUST 2026

Project Information

Project Name: TrackHire

Project Type: Full-Stack MERN Application

Development Stack

- React.js
- Node.js
- Express.js
- MongoDB
- Tailwind CSS
- TanStack Query

Architecture

- Layered Backend Architecture
- RESTful APIs
- JWT Authentication
- Service Layer Pattern
- Modular React Architecture

Repository

 **GitHub Repository**

<https://github.com/Mansi2156/TrackHire.git>

Live Demo

 **Live Application**

<https://track-hire-nine.vercel.app>

Executive Summary

Overview

TrackHire is a modern full-stack web application developed to help job seekers efficiently organize and manage their entire job search process. Instead of relying on spreadsheets, notes, and multiple applications, TrackHire centralizes job applications, interviews, resumes, companies, and progress tracking into a single platform.

The application follows a modular architecture using the MERN stack and demonstrates industry-standard software engineering practices including authentication, RESTful APIs, layered backend architecture, reusable frontend components, database relationships, and secure user data isolation.

The MVP focuses on solving the core problem of managing multiple job applications while providing a scalable foundation for future enhancements such as AI-powered resume analysis, calendar integration, notifications, and advanced analytics.

Project Highlights

- Complete MERN Stack application
 - Secure JWT Authentication
 - Resume Management
 - Company Management
 - Job Application Pipeline
 - Interview Tracking
 - Analytics Dashboard
 - RESTful API Architecture
 - MongoDB Database Design
 - Layered Backend Architecture
 - Responsive User Interface
-

Project Overview

Introduction

Searching for a job often involves managing dozens of applications simultaneously across multiple job portals and companies. As the number of applications increases, keeping track of application status, interview schedules, recruiter information, submitted resumes, and follow-up activities becomes increasingly difficult.

TrackHire addresses this challenge by providing a centralized platform where users can monitor every stage of their job search from a single dashboard.

Rather than acting as a job portal, TrackHire functions as a personal job application management system that enables users to organize their job search efficiently while providing meaningful insights into their progress.

Objectives

The primary objectives of TrackHire are:

- Centralize job search management
 - Organize application data efficiently
 - Track interview progress
 - Manage multiple resume versions
 - Maintain structured company information
 - Provide meaningful dashboard insights
 - Reduce manual tracking efforts
 - Improve overall job search productivity
-

Target Users

TrackHire is designed for:

- Students and Fresh Graduates
 - Experienced Professionals
 - Career Switchers
 - Anyone actively managing multiple job applications
-

Problem Statement

Managing a modern job search involves handling information from multiple sources.

Applicants often encounter challenges such as:

- Losing track of application status
- Forgetting interview schedules
- Managing multiple resume versions manually
- Scattered recruiter information
- Difficulty monitoring application progress

- No centralized analytics for job search performance

Traditional spreadsheets become increasingly difficult to maintain as applications grow, resulting in duplicated information, missed opportunities, and inefficient follow-up.

Solution

TrackHire provides a structured and centralized solution that simplifies the entire job search lifecycle.

The platform allows users to:

- Create and manage job applications
- Track applications through multiple hiring stages
- Schedule and manage interviews
- Upload and organize resume versions
- Store company information
- Monitor dashboard analytics
- Configure account preferences
- Securely manage personal data

Every module is integrated to provide a seamless workflow while maintaining clear separation of concerns in both the frontend and backend architecture.

Core Features

1. Authentication & User Management

TrackHire implements secure JWT-based authentication to ensure every user's data remains private and isolated.

Features

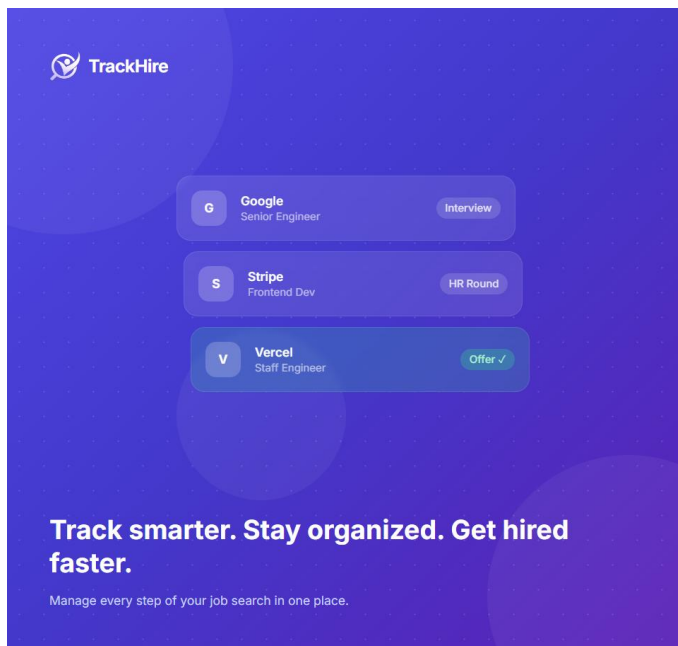
- User Registration
- Secure Login
- Password Encryption using bcrypt
- JWT Authentication
- Protected Routes
- Session Persistence

- Profile Management
- Password Change
- Account Deactivation

Technical Highlights

- Stateless authentication
- Password hashing
- Authorization middleware
- Route protection
- User ownership validation

Authentication (Login & Register Pages)



Welcome back

Sign in to continue tracking your applications

Email address

johndoe123@gmail.com

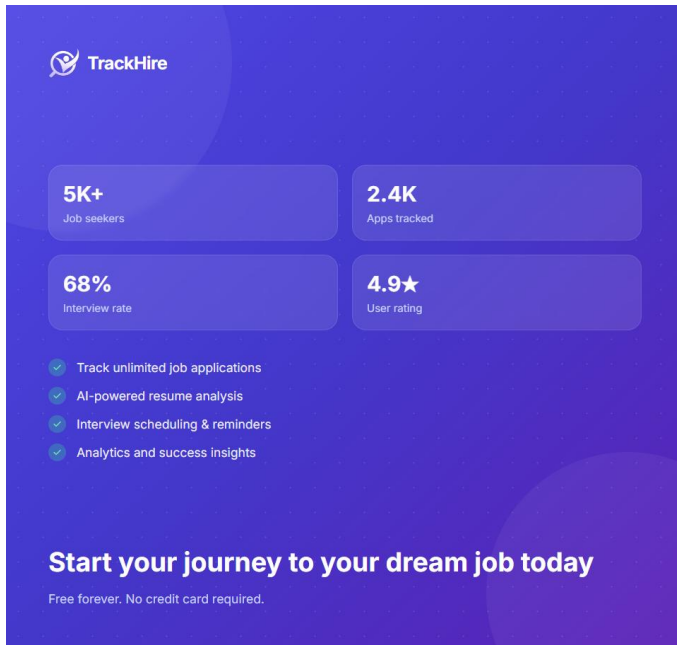
Password

.....

☒ Remember me

Sign in to TrackHire →

Don't have an account? [Create one free](#)



Create your account

Free forever. Start tracking in minutes.

Full Name

Rahul Sharma

Email address

rahul123@gmail.com

Password

.....

Confirm Password

.....

Create Free Account →

By signing up, you agree to our [Terms of Service](#) and [Privacy Policy](#)

Already have an account? [Sign in](#)

2. Dashboard & Analytics

The dashboard serves as the central overview of the user's job search activities.

Instead of displaying static information, the dashboard aggregates real-time data from multiple modules to provide meaningful insights.

Dashboard Components

- Total Applications
- Active Applications
- Upcoming Interviews
- Total Offers
- Monthly Progress
- Application Funnel
- Status Distribution
- Recent Activity
- Upcoming Deadlines

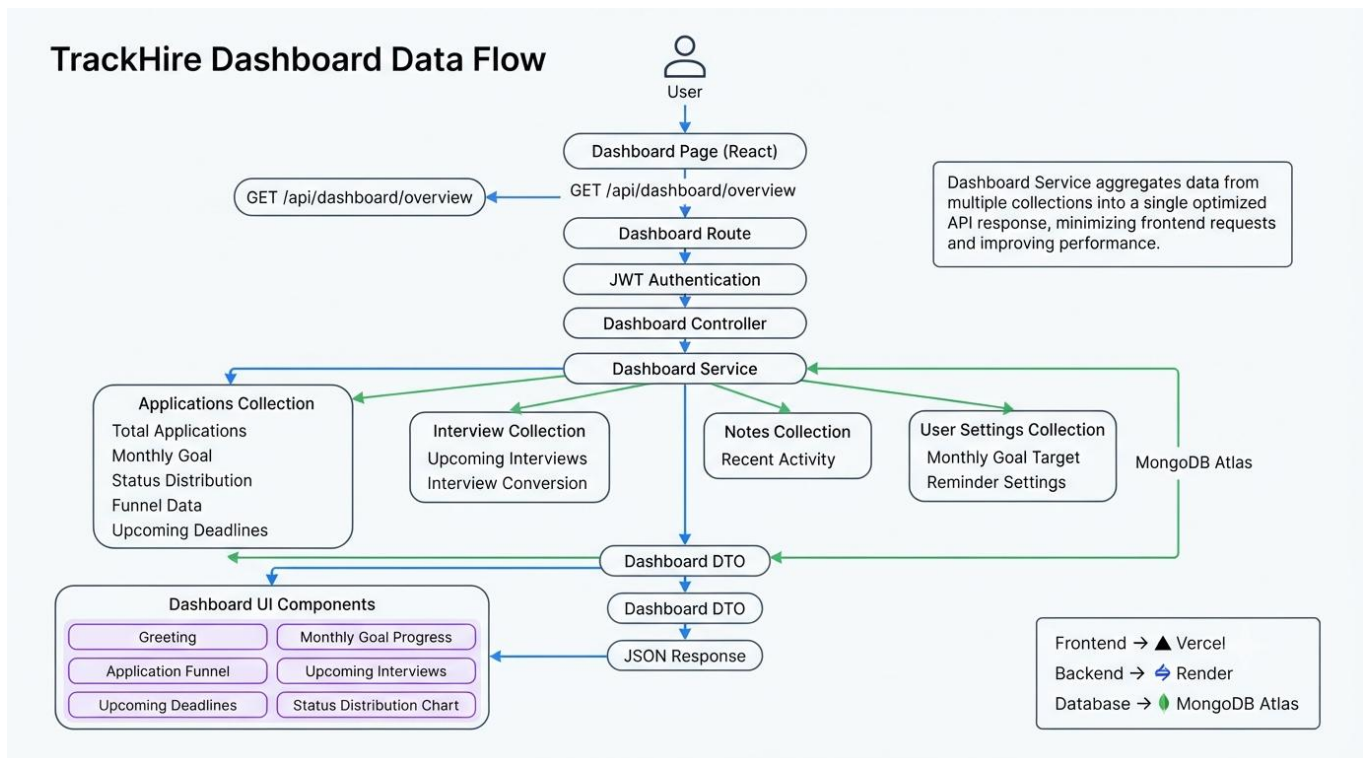
The dashboard helps users quickly understand the current state of their job search without navigating through multiple pages.

Key Benefits

- Quick overview of progress
- Visual hiring pipeline

- ## Dashboard Overview

Dashboard Flow Diagram



3. Job Application Management

The Job Application module is the core functionality of TrackHire.

It enables users to manage every application from initial submission until the final hiring decision.

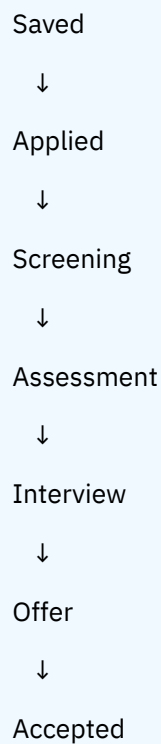
Capabilities

- Create Applications
- View Applications
- Update Information
- Delete Applications
- Archive Applications
- Search Applications
- Filter by Status
- Sorting
- Pagination

Each application stores important information including company details, position, recruiter information, work mode, job type, salary, notes, resume reference, deadlines, and interview schedules.

Application Pipeline

Applications move through a structured hiring workflow:



Alternative Outcomes:

Rejected
Withdrawn
Closed

This visual pipeline provides users with a clear understanding of where each application currently stands.

Key Features

- Structured hiring stages
- Resume linking
- Company linking
- Recruiter information
- Deadline tracking
- Notes management
- Archive support
- Duplicate prevention

- Status updates

Benefits

- Organized application tracking
- Better visibility into hiring progress
- Reduced manual effort
- Improved follow-up management

Applications List

TrackHire

Dashboard

Applications

Companies

Interviews

Resumes

Settings

J

John Doe

john.doe123@gmail.com

Logout

Applications

9 applications

+ Add Application

Q

Search by company or role...

All statuses

Newest

Active

Archived

All

☐	Company	Role	Applied	Status	Interview
☐	L LTIMindtree	Associate Software Engineer	Aug 5 Applied 1 day ago	Applied	—
☐	K KPMG	MERN stack developer	Aug 5 Applied 1 day ago	Applied	—
☐	D Deloitte	Associate Software Engineer	Aug 4 Applied 2 days ago	Interview	Aug 6
☐	Q Qualcomm	Associate Software Engineer	Aug 3 Applied 3 days ago	Applied	—
☐	Q Qualcomm	Full Stack developer	Aug 3 Applied 3 days ago	Interview	Aug 5
☐	S Sysfort	Full Stack developer	Jul 26 Applied 11 days ago	Applied	—

Add/Edit Application Form

Add New Application

Track a new job opportunity

Company Name *

KPMG

Role / Job Title *

MERN stack developer

Work Mode

On-site

Location *

Pune

Employment Type

Full-Time

Salary Range

9 - 12 LPA

Application Date *

05-08-2026

Interview Date

dd-mm-yyyy

Application Deadline

10-08-2026

Recruiter Name

Rahul Mishra

Recruiter Email

reach.rahulmishra45@gmail.com

Job Description

Proficiency in MongoDB, Express.js, React.js, and Node.js languages. Strong command of modern JavaScript (ES6+) and TypeScript (often preferred). Tools & Workflow: Experience with Git/GitHub, REST/GraphQL APIs, and package managers like npm or yarn. Education: Bachelor's degree in Computer Science, IT, or equivalent practical experience

Application URL

<https://www.instahyre.com/job-392259-mern-stack-developer-at-kpmg-2-work-from-ho>

Current Status *

Saved

Applied

Screening

Assessment

Interview

Offer

Accepted

Rejected

Withdrawn

Closed

Resume

MERN stack (v1)

Notes

Any notes, referrals, or reminders...

Save Application

Cancel

Application Details Page

[← Back to Applications](#)**Qualcomm**

Full Stack developer

Interview



APPLIED

Aug 3, 2026



LOCATION

Mumbai



EMPLOYMENT

Full-Time



SALARY

9 - 7 LPA

Recruiter Information

LS

Lisa Seren

lisa@qualc.com

Email

Job Description[View posting](#)

A full stack developer builds and maintains complete web applications, handling both the client-side user interface and the server-side logic. You can view a sample template on Workable's Job Description or read a detailed overview on Indeed's Job Description Guide.

Key Responsibilities

Front-End Development: Design and build user interfaces using HTML, CSS, and JavaScript frameworks (such as React, Angular, or Vue).

Back-End Development: Create server-side logic, maintain databases, and write robust APIs using languages like Node.js, Python, Java, or Ruby.

Testing and Debugging: Test software for responsiveness, fix bugs, and optimize application performance.

Deployment: Manage the end-to-end life cycle of applications from conception to launch.

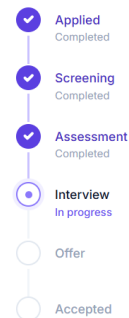
Required Skills and Qualifications

Languages: Proficiency in HTML, CSS, JavaScript, and a server language (Python, Java, PHP, or .NET).

Databases: Experience with SQL or NoSQL database management (MySQL, MongoDB, PostgreSQL).

Tools: Familiarity with Git version control and cloud environments.

Soft Skills: Strong problem-solving abilities and teamwork.

Application Timeline**Details**

Resume Used

Full Stack (v1)

Next Interview

Aug 5, 2026

Interviews[+ Schedule Interview](#)Round 1 - Technical
August 3, 2026

Passed

HR Round - HR
August 5, 2026

Scheduled

Round 2 - Technical
August 5, 2026

Passed

4. Resume Management

The Resume Management module allows users to securely organize and maintain multiple resume versions throughout their job search.

Instead of storing resumes in different folders with inconsistent naming conventions, TrackHire provides a centralized resume library with version tracking and direct integration with job applications.

Features

- Upload PDF and DOCX resumes
- Resume version management
- Resume tagging
- Default resume selection

- Resume preview
- Resume download/replace/rename
- Link resumes directly with applications
- Environment-based file storage (Local file storage for development & Cloudinary storage for production)

Each uploaded resume can be assigned a meaningful title and tags, making it easy to organize resumes for different technologies or job roles.

Key Benefits

- Organized resume library
- Easy version tracking
- Faster application process
- Better resume organization
- Direct application integration

Resume Management Page


TrackHire

Dashboard

Applications

Companies

Interviews

Resumes

Settings

John Doe

johndoe123@gmail.com

Logout

Resume Manager

3 resumes · Track which resume gets results

Upload Resume

Total Resumes

3

3 uploaded

Storage Used

390 KB

Across all resumes

Default Resume

Full Stack

v1

Last Updated

Aug 6

Full Stack

PDF

Full Stack

v2 · 204 KB

Updated Aug 6 ·

Used in 0 Applications

React

Node

ExpressJS

MySql

☆

↓

✎

🗑

PDF

★ Default

Full Stack

v1 · 93 KB

Updated Jul 31 ·

Used in 2 Applications

React

Node

PostgreSQL

↓

✎

🗑

PDF

Software Engineer

v1 · 93 KB

Updated Jul 24 ·

Used in 5 Applications

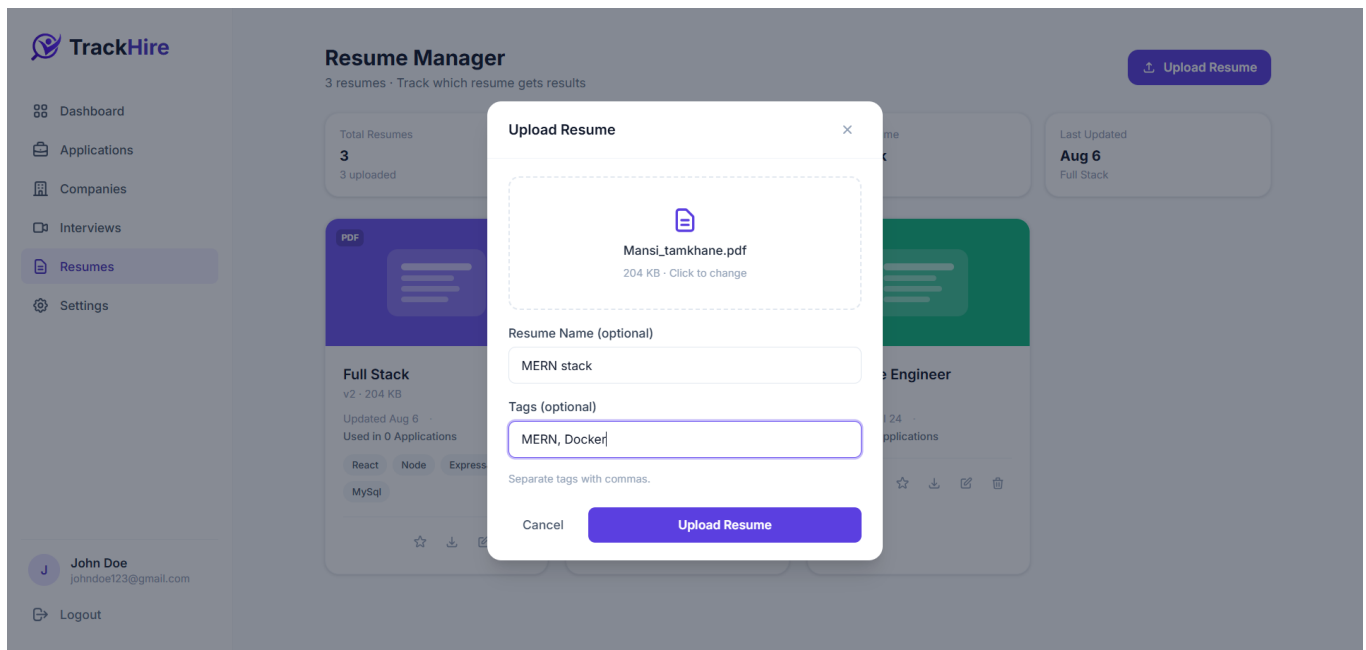
☆

↓

✎

🗑

Upload Resume Dialog



5. Company Management

TrackHire maintains structured company profiles that eliminate duplicate information across applications and provide better organization during the job search.

Instead of repeatedly entering company details for every application, users can create reusable company profiles containing important business information.

Features

- Create Company Profiles
- Update Company Information
- Delete Companies
- Company Search
- Company Statistics
- Website Information
- Industry Classification
- Company Notes
- Linked Applications

Each company profile automatically connects with related job applications, allowing users to quickly understand their history with a particular organization.

Benefits

- Centralized company information
- Better organization
- Reduced duplicate data
- Company-wise application tracking
- Improved search experience

Companies List

TrackHire

Dashboard

Applications

Companies

Interviews

Resumes

Settings

John Doe

johndoe123@gmail.com

Logout

Companies

Track companies you've applied to and manage relationships

+ Add Company

TOTAL

10

Total Companies

ACTIVE

10

Active Applications

INTERVIEWS

9

Interviews

OFFERS

0

Offers

Search companies...

All Industries

Most Recent

Company	Industry	Location	Website	Applications	Last Activity
M Mastercard	Fintech	Pune	www.mastercard.com/in/en.html	0 apps 0 interviews	Today
A Accenture	Productivity SaaS	Pune	www.accenture.com/in-en	1 apps 2 interviews	Today
M Mphasis	Productivity SaaS	Bengaluru	www.mphasis.com/home.html	1 apps 1 interviews	Today
C Capgemini	Technology	Mumbai	www.capgemini.com/	0 apps 0 interviews	Today
L LTIMindtree	Technology	Mumbai	www.ltm.com/	1 apps 0 interviews	Today
H HCLTech	Technology	Noida	www.hcltech.com/	1 apps 1 interviews	Today
T TCS	Technology	Mumbai	www.tcs.com/	1 apps 1 interviews	Today
K KPMG	Productivity SaaS	Pune	kpmg.com/in/en.html	2 apps 0 interviews	Today
Q Qualcomm	Productivity SaaS	Mumbai	www.qualcomm.com/	2 apps 3 interviews	Today
D Deloitte	Productivity SaaS	Pune	www.deloitte.com/in/en.html	2 apps 1 interviews	2d ago

Company Details Page

[← Back to Companies](#)

Deloitte

Productivity SaaS @ Pune www.deloitte.com/in/en.html
Last activity 2d ago

**2**

Total Applications

2

Active

1

Interviews

0

Offers

0

Rejections

About

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Notes

Big 4 company

Applications at Deloitte

[+ Add Application](#)

Role	Applied	Status	Interview
Software Engineer	—	Saved	—
Associate Software Engineer	Aug 4	Interview	Aug 6

[← Back to Companies](#)

Add New Company

Save a company you want to track



Persistent Systems

Technology · Pune

Company Name *

Persistent Systems

Website

<https://www.persistent.com/>

Industry

Technology

Location

Pune

Description

Persistent Systems is a global technology services and digital engineering company founded in 1990 by Dr. Anand Deshpande. Headquartered in Pune, India, the public company provides AI-driven software engineering, cloud computing, and enterprise modernization to major global industries. Company Overview & Leadership Founded: 1990 in Pune, India, by Anand Deshpande, CEO. Sandeep Kalia serves as the Executive

Notes

- Q1 FY27 Revenue: Achieved \$452.4M, up 16.1% year-on-year and 3.8% sequentially.
- Total employee headcount rose by 4.1% sequentially to 28,640.

Add Company

× Cancel

6. Interview Management

Interview Management helps users organize every interview round throughout the hiring process.

Users can schedule interviews, record interview outcomes, maintain preparation notes, and track progress from the initial screening to the final offer.

Features

- Schedule Interviews
- Multiple Interview Rounds
- Interview Types
- Interview Modes
- Preparation Notes
- Feedback Notes
- Interview Status Updates
- Application Integration

Each interview is linked directly with its corresponding job application, creating a complete hiring timeline.

Supported Interview Types

- HR Interview
- Technical Interview
- Behavioral Interview
- System Design
- Panel Interview
- Take Home Assignment
- Final Interview

Key Benefits

- Organized interview schedule
- Better preparation
- Interview history
- Round-wise progress tracking
- Improved follow-up

[← Back to Interviews](#)

Schedule Interview

Add a new interview to your tracker

Job Application *

Qualcomm — Full Stack developer

Qualcomm · Full Stack developer · Interview

Interview Round *

HR Round

Interview Type *

HR

Interview Date *

05-08-2026

Interview Time *

11:30

Interview Mode *

Video Call

Phone

On-site

Async

Meeting Link

https://meet.google.com/abc-defg-hij

Preparation Notes

- Prepare a 60-sec self-introduction
- Research the company's products, services, and values,
- Practice clear communication and maintain a positive attitude.

Feedback & Result

How did it go? Add feedback and outcome notes after the interview...

Status

Scheduled

Completed

Passed

Failed

Cancelled

Rescheduled

Schedule Interview

× Cancel

7. User Settings

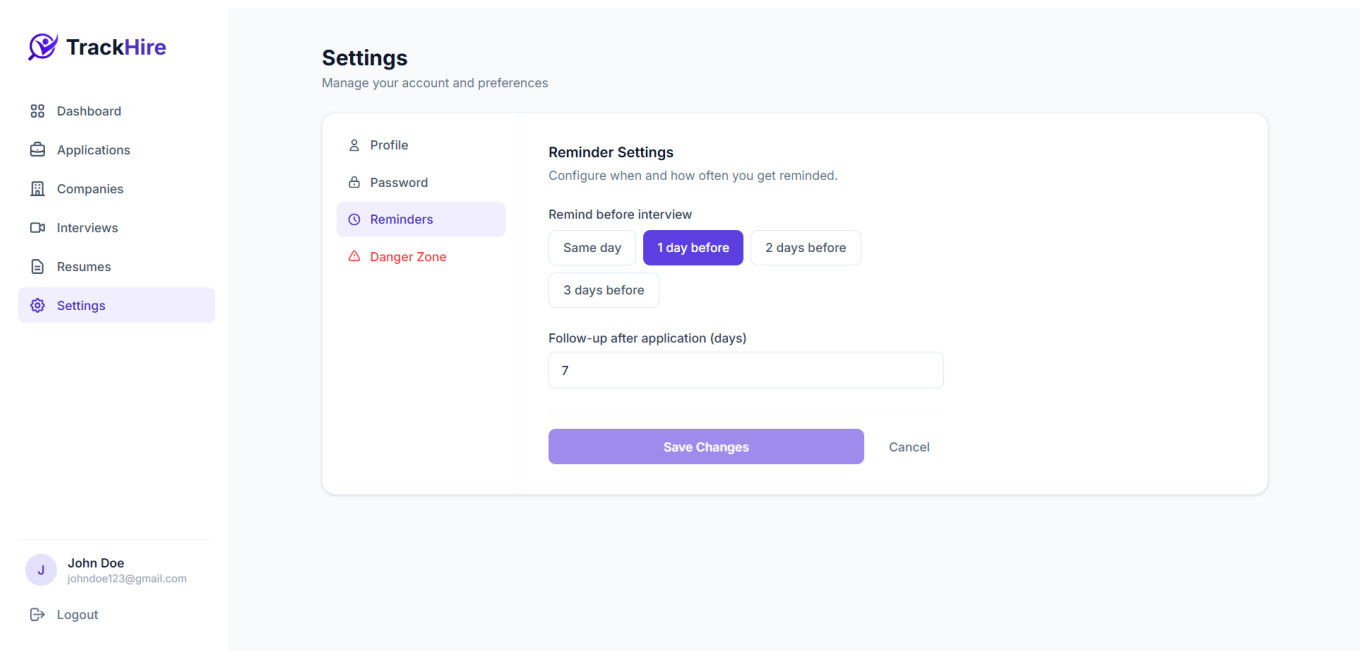
The Settings module enables users to manage their personal profile and application preferences while maintaining account security.

Available Settings

Profile

Users can update:

- Full Name
- Job Title
- Location
- Bio



Technology Stack

TrackHire is developed using the MERN stack with modern frontend and backend technologies.

Layer	Technology
Frontend	React.js
Build Tool	Vite
Styling	Tailwind CSS
Routing	React Router DOM
Forms	React Hook Form
State Management	TanStack Query
Charts	Recharts
HTTP Client	Axios
Backend	Node.js
Framework	Express.js
Database	MongoDB Atlas
ODM	Mongoose
Authentication	JWT

Layer	Technology
Password Encryption	bcrypt
Validation	express-validator
File Upload	Multer + Cloudinary

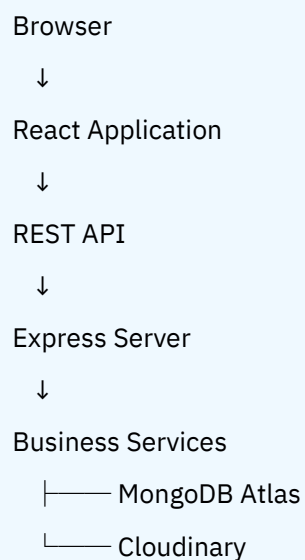
Development Tools

- ESLint
- Nodemon
- Concurrently
- Git
- GitHub

The selected technology stack provides a scalable, maintainable, and production-ready foundation while enabling rapid frontend development and efficient backend API implementation.

System Architecture

TrackHire follows a layered architecture that separates responsibilities between presentation, business logic, and data access.



This separation improves maintainability, scalability, and testability while keeping the codebase organized.

Architecture Layers

1. Presentation Layer

Responsible for rendering the user interface and handling user interactions.

Major responsibilities:

- Routing
- Forms
- UI Components
- Dashboard
- Charts
- Client Validation

2. API Layer

Acts as the communication bridge between the frontend and backend.

Responsibilities include:

- Request handling
- Authentication
- Validation
- Error handling
- Response formatting

3. Business Layer

Contains the application's core business logic.

Responsibilities include:

- CRUD operations
- Ownership validation
- Dashboard calculations
- Resume management
- Company linking

- Interview management
-

4. Data Layer

Responsible for interacting with MongoDB using Mongoose.

Responsibilities include:

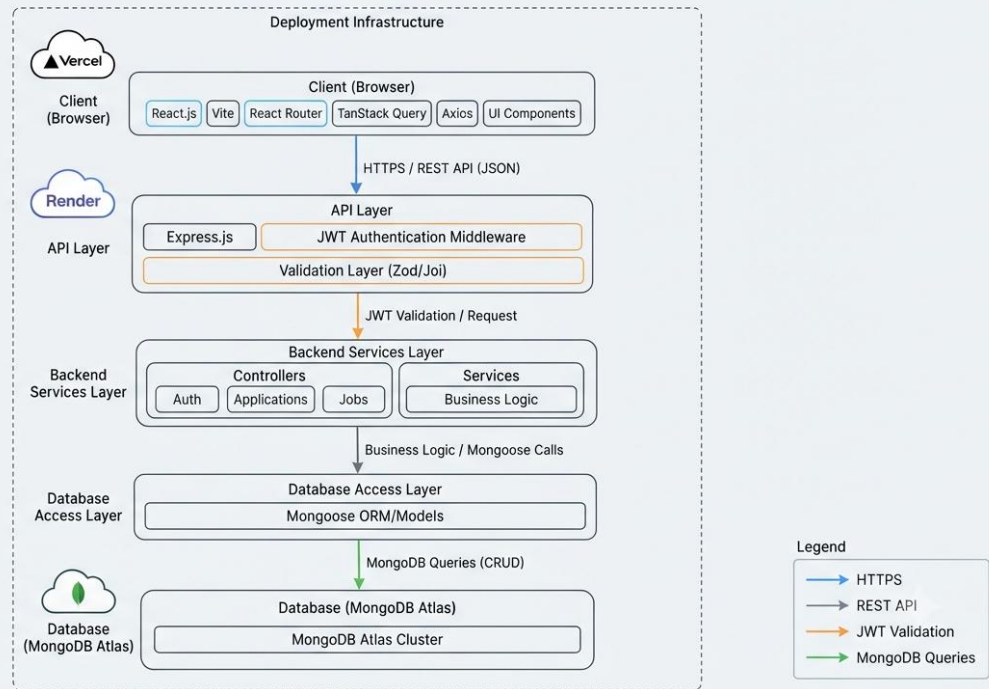
- Data persistence
 - Relationships
 - Validation
 - Query execution
 - Indexing
-

Architectural Highlights

- Layered Architecture
- RESTful APIs
- JWT Authentication
- Modular Code Structure
- Service Layer Pattern
- Database Abstraction
- Reusable Components
- Secure User Isolation

System Architecture Diagram

TrackHire



Why This Architecture?

The layered architecture improves maintainability by separating concerns across different modules.

Each layer has a single responsibility, making the application easier to understand, test, and extend as new features are introduced.

This design also enables future enhancements without requiring major architectural changes.

Database Design

TrackHire uses **MongoDB Atlas** as its primary database, with **Mongoose** as the Object Data Modeling (ODM) library. The database is designed around the core entities involved in a job search workflow while maintaining clear relationships between modules.

The application follows a document-oriented data model, providing flexibility for storing structured information such as job applications, interview records, resumes, and company profiles.

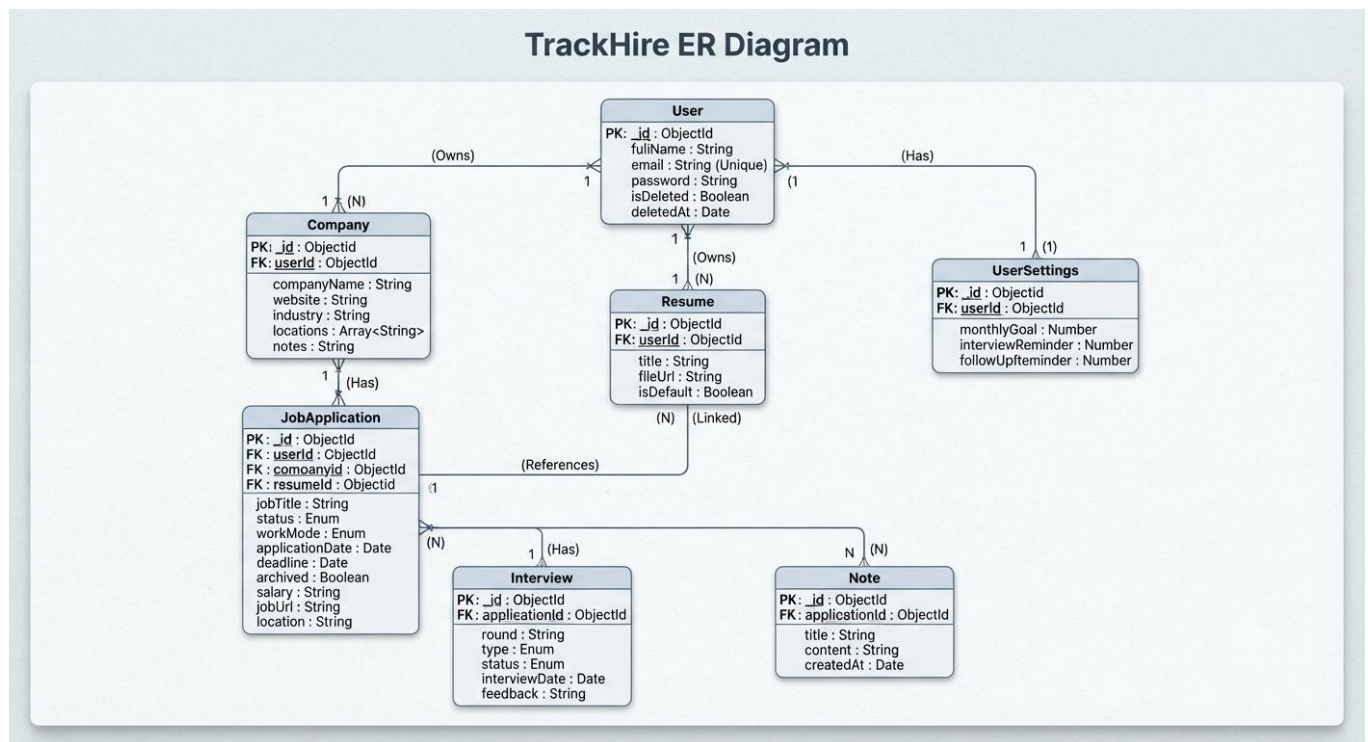
Core Collections

The MVP consists of five primary collections:

Collection	Purpose
Users	Stores user accounts, profile information, reminder preferences, and authentication data.
Job Applications	Stores job application details, hiring status, recruiter information, and linked resources.
Companies	Maintains reusable company profiles and related application statistics.
Resumes	Stores resume metadata, version information, storage references, and file details.
Interviews	Tracks interview schedules, rounds, outcomes, and preparation notes.

The relationships between these collections enable users to organize their complete job search without duplicating information.

Entity Relationship (ER) Diagram



Database Design Highlights

The database was designed with scalability and maintainability in mind.

Key design considerations include:

- User-based data isolation

- Structured document relationships
- Compound indexing for frequently accessed data
- Resume and company references
- Soft-delete support for user accounts
- Automatic timestamps for auditing

These design choices improve query performance while keeping the data model easy to extend in future releases.

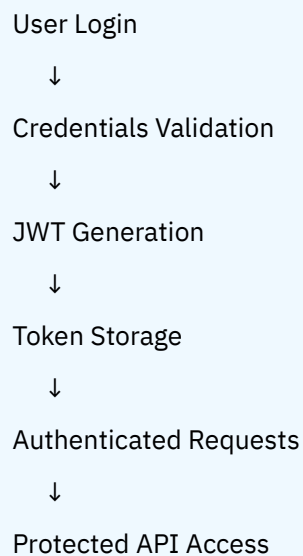
Authentication & Security

Security is a core aspect of TrackHire. Every authenticated request is verified before accessing protected resources.

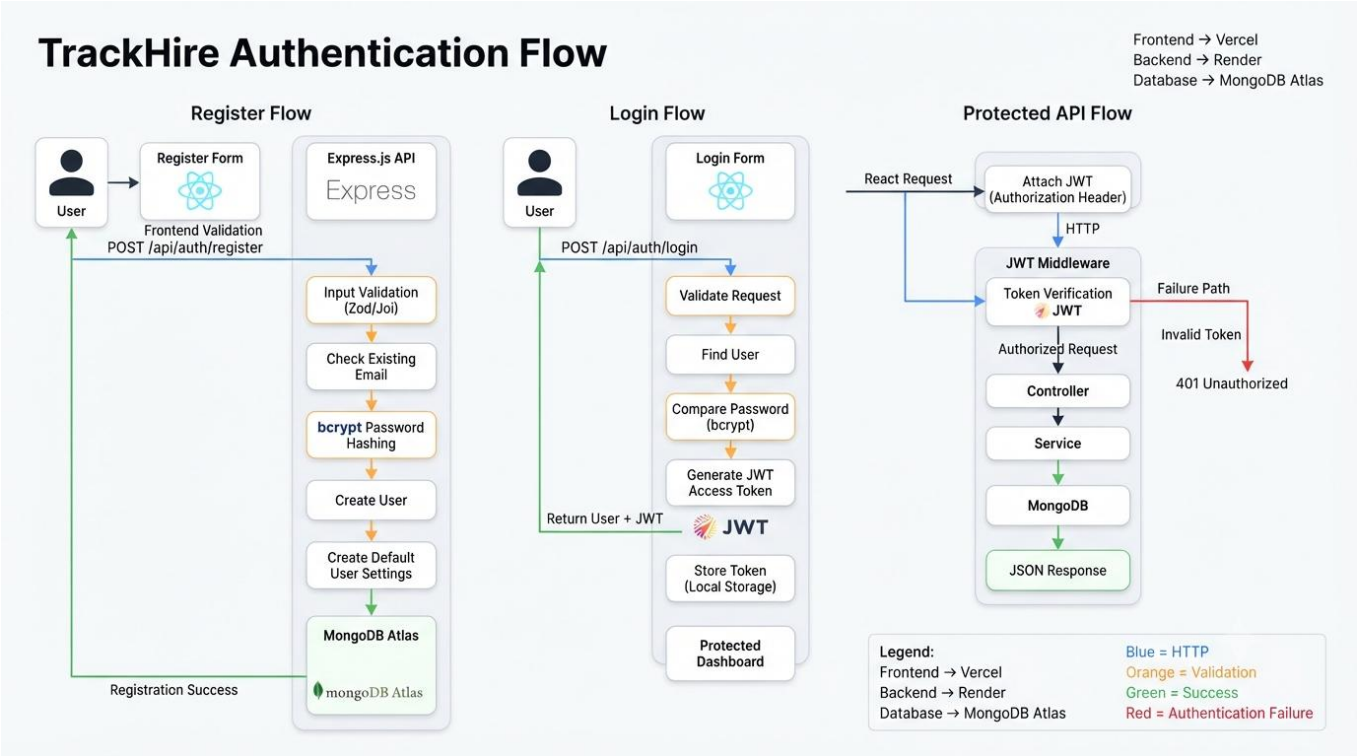
The application implements JWT-based authentication, ensuring that users can only access their own data throughout the platform.

Authentication Flow

The authentication process follows a straightforward and secure workflow:



Authentication Flow Diagram



Security Features

Secure Authentication

- JWT-based authentication
- Password hashing using bcrypt
- Protected frontend routes
- Protected backend APIs

Authorization

Every API request validates the authenticated user before performing any database operation.

This ensures:

- Users can only access their own resources.
- Cross-user access is prevented.
- Data ownership remains secure.

Request Validation

All incoming requests are validated before business logic is executed.

Validation includes:

- Required fields
 - Email validation
 - Enum validation
 - File validation
 - Input sanitization
 - Business rule validation
-

Account Protection

The application also implements:

- Secure password updates
- Session persistence
- Soft account deletion
- Unauthorized access prevention

These mechanisms provide a secure and reliable user experience while keeping the authentication system lightweight and scalable.

Application Workflow

TrackHire follows a structured workflow that mirrors the real-world hiring process.

Users can create applications, schedule interviews, manage resumes, and monitor progress through a centralized dashboard.

Overall Workflow

Register/Login



Dashboard



Create Company



Upload Resume



Create Job Application



Track Hiring Stages

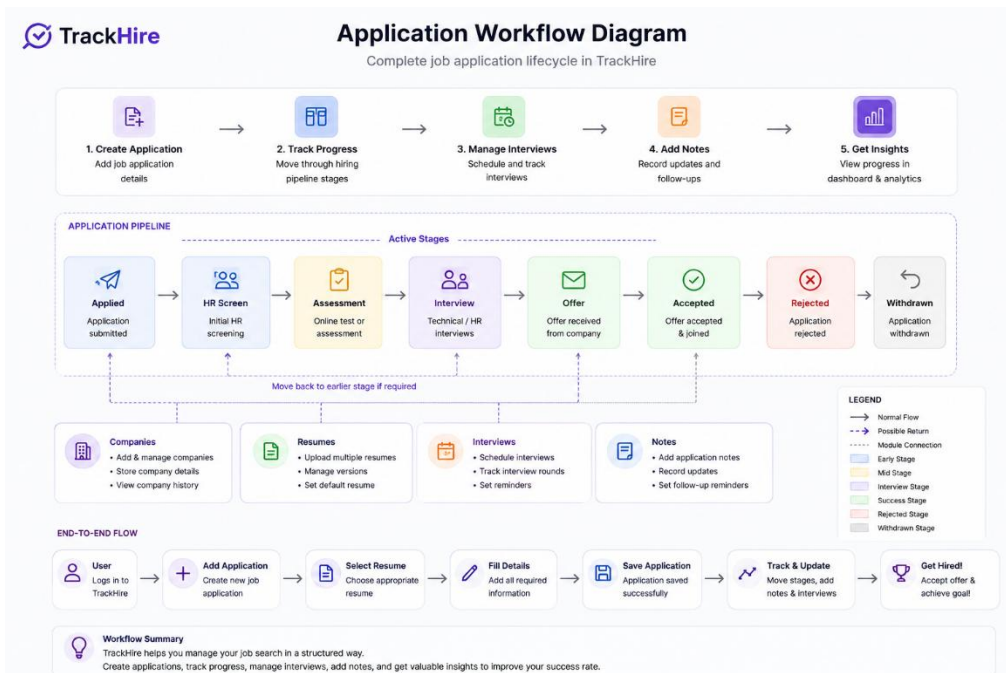


Schedule Interviews



Receive Offer / Close Application

Application Workflow Diagram



Application Lifecycle

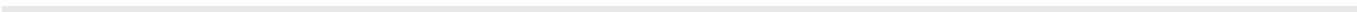
Each application progresses through a structured hiring pipeline.



Alternative outcomes include:

- Rejected
- Withdrawn
- Closed

This workflow provides users with complete visibility into their hiring progress.



API Overview

TrackHire follows RESTful API principles for communication between the frontend and backend.

Each module exposes dedicated endpoints for managing application data while maintaining consistent request and response structures.

API Modules

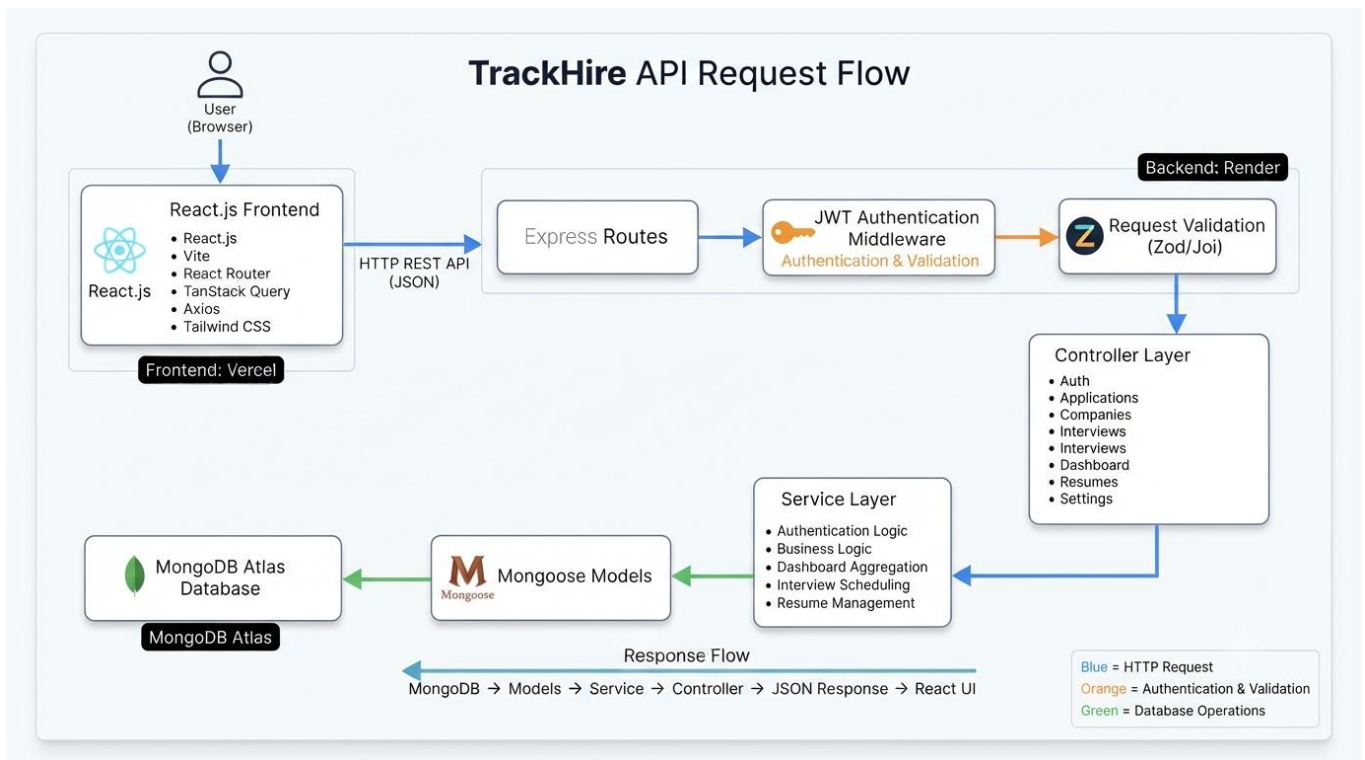
Module	Responsibility
Authentication	User registration, login, authentication

Module	Responsibility
Dashboard	Analytics and aggregated metrics
Applications	CRUD operations for job applications
Companies	Company management
Resumes	Resume upload and management
Interviews	Interview scheduling and tracking
User Settings	Profile and account management

API Characteristics

- RESTful architecture
- JSON request/response format
- JWT Authorization
- Consistent error handling
- Request validation
- Pagination support
- Filtering and searching
- Status-based operations

API Request Flow Diagram



Project Structure

TrackHire follows a modular folder structure that separates frontend and backend responsibilities. This organization improves maintainability and allows each module to evolve independently.

Frontend

```

client/
|
|—— components/
|—— pages/
|—— layouts/
|—— hooks/
|—— context/
|—— api/
|—— utils/
|—— constants/

```

Responsibilities

- User Interface
- Routing

- API Communication
 - State Management
 - Form Handling
 - Reusable Components
-

Backend

```
server/  
|  
├── routes/  
├── controllers/  
├── services/  
├── models/  
├── middleware/  
├── validators/  
├── config/  
├── utils/  
└── constants/
```

Responsibilities

- REST APIs
 - Authentication
 - Business Logic
 - Database Access
 - Validation
 - File Upload & Storage
 - Error Handling
-

Project Design Principles

The application is built around several software engineering principles:

- Separation of Concerns
- Single Responsibility Principle
- Modular Architecture
- Reusable Components
- Secure Data Access

- Scalable Folder Structure
- Maintainable Codebase

These principles make the project easier to understand, extend, and maintain as additional features are introduced.

Engineering Decisions

During development, several architectural decisions were made to improve scalability, maintainability, and long-term extensibility rather than focusing only on implementing features.

1. Layered Backend Architecture

The backend follows a layered architecture where each layer has a single responsibility.

```
graph TD; Routes --> Controllers; Controllers --> Services; Services --> Models; Models --> Database;
```

Routes
↓
Controllers
↓
Services
↓
Models
↓
Database

Benefits

- Clear separation of concerns
 - Easier debugging
 - Better maintainability
 - Reusable business logic
 - Improved scalability
-

2. Service Layer Pattern

Instead of placing business logic inside controllers, TrackHire delegates all application logic to dedicated service classes.

This approach keeps controllers lightweight while making business logic reusable, testable, and easier to maintain.

3. JWT Authentication

A stateless JWT authentication mechanism was selected because it provides a simple and scalable authentication strategy for modern web applications.

Benefits include:

- Stateless authentication
- Simplified frontend integration
- Secure route protection
- Easy API authorization

4. MongoDB Document Model

MongoDB was selected because the application's data naturally fits a document-based structure.

This allows flexible schema evolution while maintaining strong relationships between users, applications, companies, resumes, and interviews.

5. Modular Frontend Design

The frontend is built using reusable React components and custom hooks.

Benefits include:

- Code reuse
- Easier maintenance
- Consistent UI
- Simplified feature expansion

Challenges & Solutions

Developing TrackHire involved solving several real-world engineering challenges beyond standard CRUD operations.

Challenge 1

Managing Multiple Related Modules

The application contains interconnected modules such as Applications, Companies, Interviews, and Resumes.

Maintaining relationships between these entities while keeping the architecture modular required careful database design.

Solution

Entity relationships were established using MongoDB references while keeping each module independently maintainable.

Challenge 2

Secure User Data Isolation

Every authenticated user must only access their own resources.

Solution

JWT authentication combined with ownership validation ensures that every database query is scoped to the authenticated user.

Challenge 3

Efficient Dashboard Aggregation

The dashboard requires data from multiple collections.

Fetching every metric separately would increase API requests and reduce performance.

Solution

A dedicated dashboard service aggregates all required information and returns it through a single optimized API response.

Challenge 4

Resume Version Management

Users frequently maintain multiple versions of resumes.

Managing uploads without confusion required version control.

Solution

Resume metadata is stored in MongoDB while file storage is environment-dependent. Development uses local server storage, while production uses Cloudinary for persistent file storage. Resume versioning, default resume selection, tagging, preview, download, replacement, and deletion are supported.

Performance & Optimizations

Several optimization techniques were implemented throughout the application to improve responsiveness and scalability.

Backend Optimizations

- Compound MongoDB indexes
 - Optimized database queries
 - Centralized business logic
 - Efficient aggregation pipelines
 - Validation before database operations
-

Frontend Optimizations

- TanStack Query caching
 - API cache invalidation
 - Reusable components
 - Lazy data loading
 - Efficient form handling
-

Application Performance

Performance improvements include:

- Reduced redundant API requests
 - Faster dashboard loading
 - Optimized database lookups
 - Improved user experience
 - Better scalability for future modules
-

Future Scope

The current implementation represents the completed MVP.

Future releases will extend the platform with additional productivity and intelligence features.

Planned Enhancements

1. Analytics

- Advanced analytics dashboard
 - Historical trends
 - Monthly reports
 - Success rate analysis
-

2. Notifications

- Interview reminders
 - Follow-up reminders
 - Email notifications
 - Browser notifications
-

3. Calendar Integration

- Google Calendar
 - Outlook Calendar
 - Automatic interview scheduling
-

4. AI Features

- Resume analysis
 - Resume tailoring suggestions
 - Job description matching
 - Interview preparation assistance
-

5. Additional Features

- Global Search
- Advanced Filtering
- Activity Timeline

- Mobile Application
 - Team Collaboration
-

Key Learnings

TrackHire provided practical experience in designing and developing a production-style full-stack application.

The project strengthened understanding of:

- Full-stack application architecture
- RESTful API development
- MongoDB data modeling
- Authentication and authorization
- React application architecture
- Backend service-layer design
- Component-based frontend development
- Secure file handling and cloud storage integration
- Performance optimization
- Software engineering best practices

More importantly, the project emphasized designing maintainable systems rather than simply implementing CRUD functionality.

Conclusion

TrackHire demonstrates the implementation of a modern full-stack web application built using the MERN stack. The project successfully delivers a centralized platform for managing job applications, interviews, resumes, and companies while maintaining a clean architecture, secure authentication, modular codebase, and scalable backend design.

The completed MVP showcases industry-standard development practices including layered architecture, RESTful APIs, JWT authentication, reusable React components, optimized database design, and responsive user experience. Beyond the delivered functionality, the project has been designed with extensibility in mind, making it straightforward to introduce advanced capabilities such as AI-powered insights, calendar integration, notifications, and enhanced analytics in future releases.

TrackHire reflects not only the implementation of application features but also thoughtful engineering decisions focused on maintainability, scalability, and real-world software development practices.